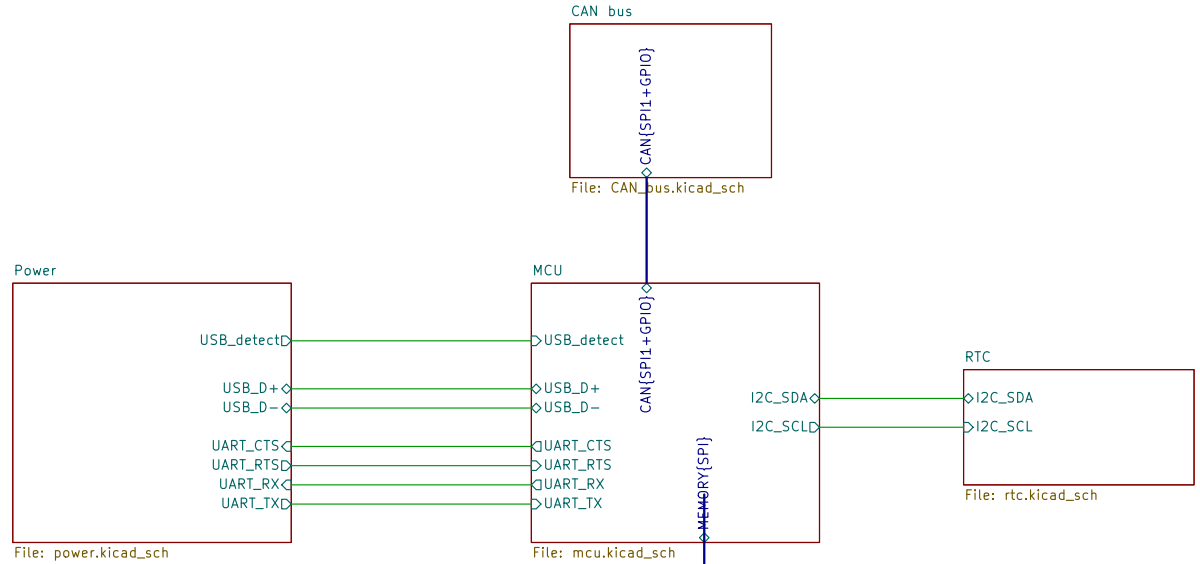


H101
MountingHole

H102
MountingHole

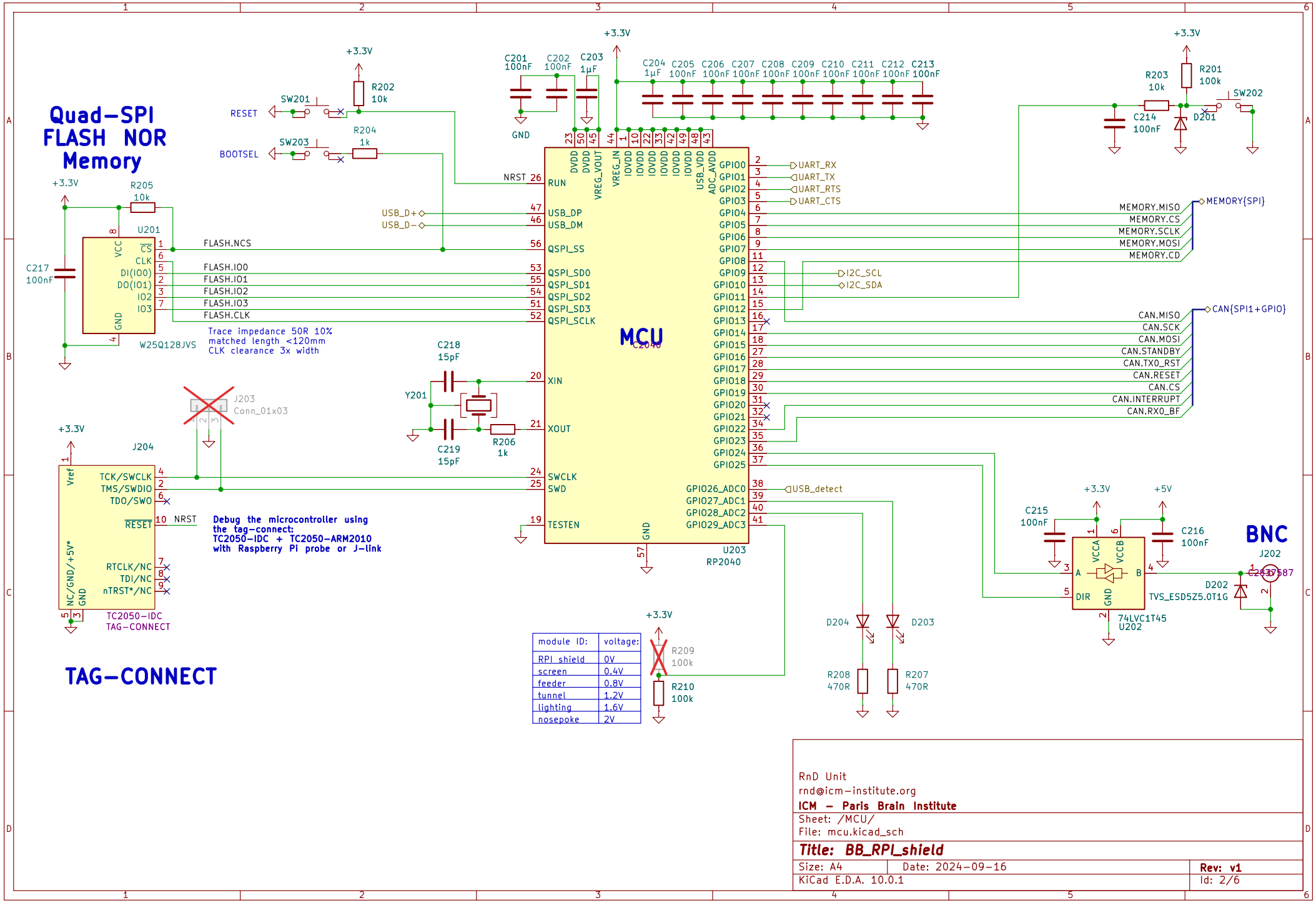
H103
MountingHole

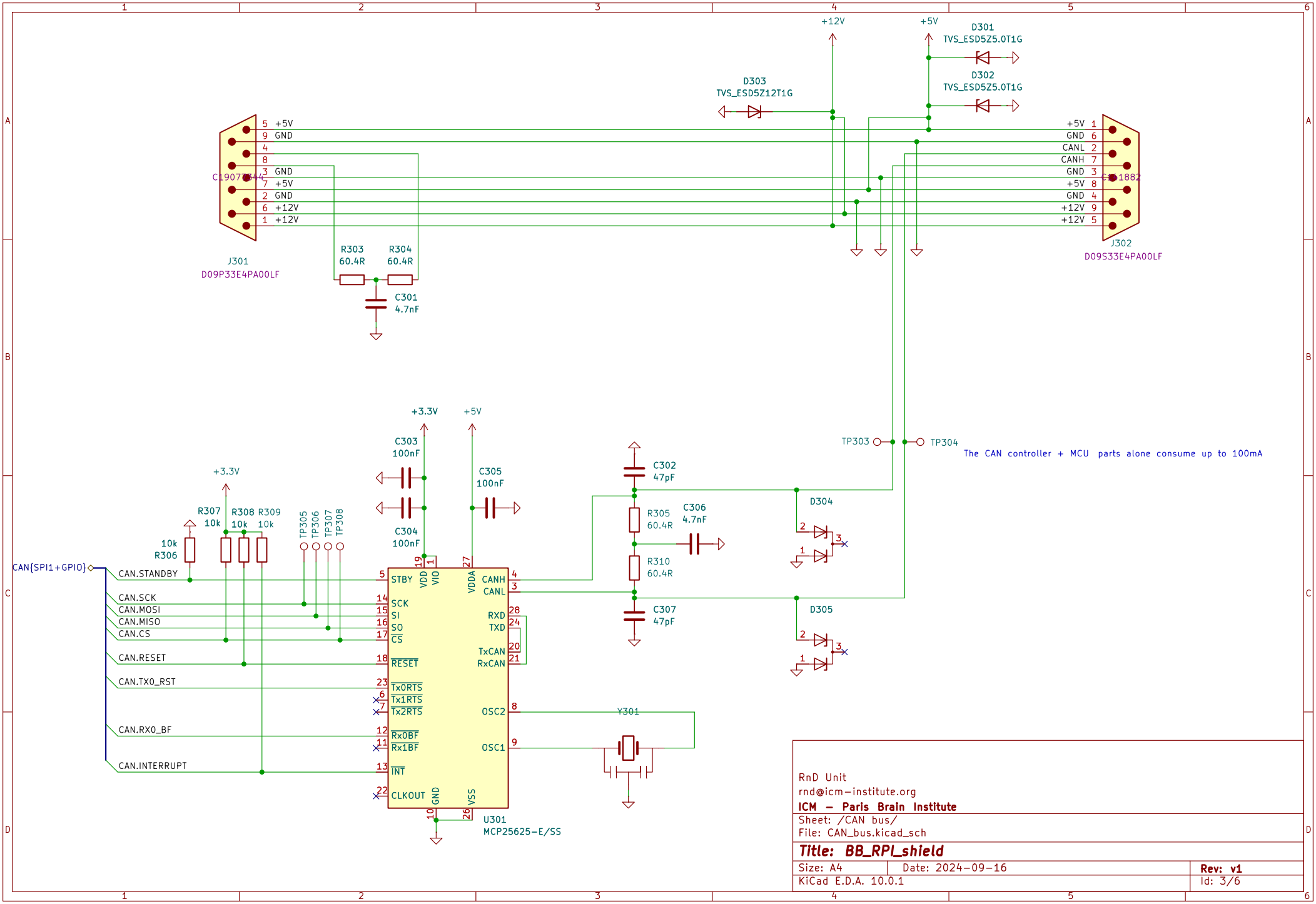
RnD Unit
rnd@icm-institute.org
ICM – Paris Brain Institute

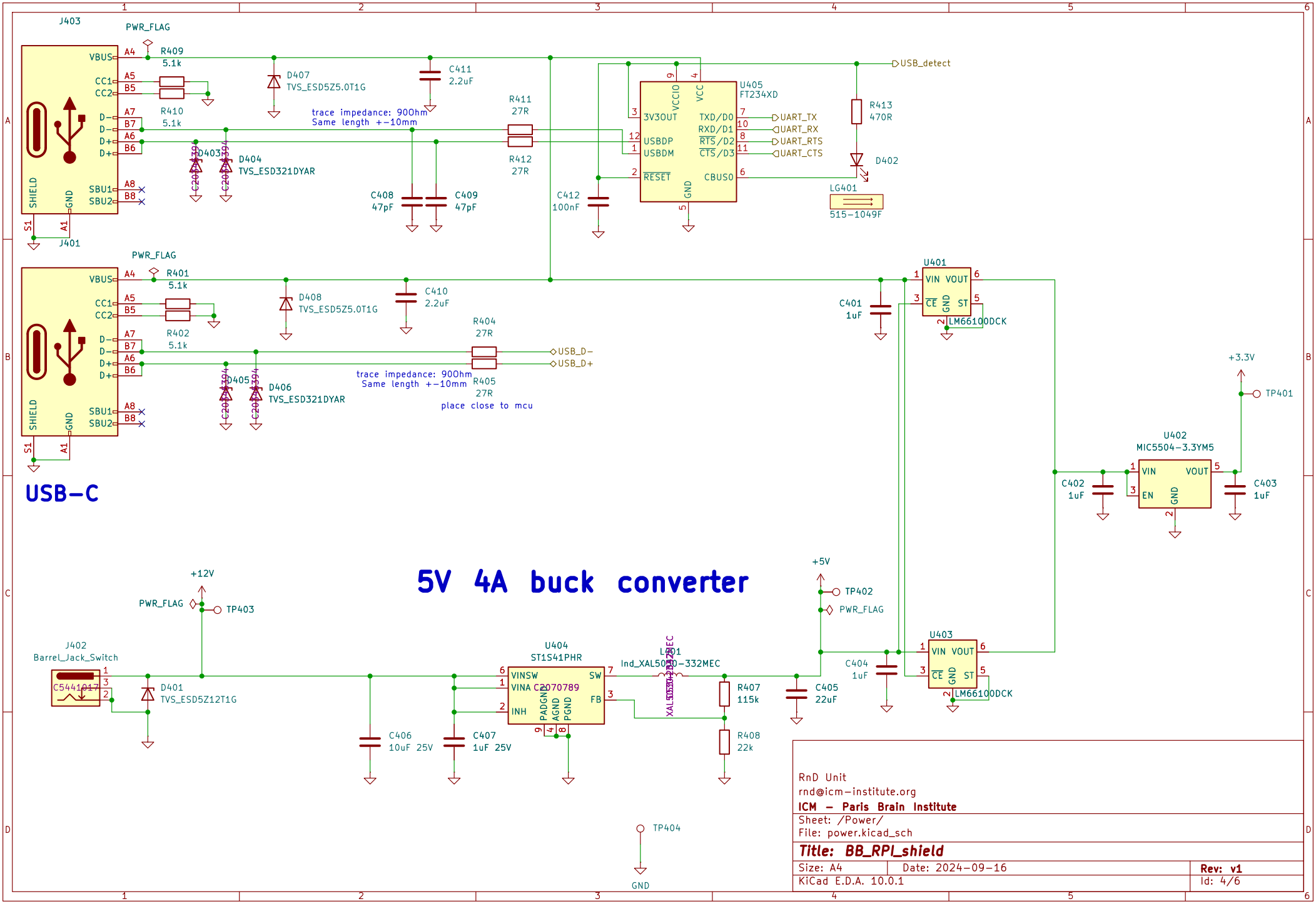
Sheet: /
File: BB_RPI_shield_V1.kicad_sch

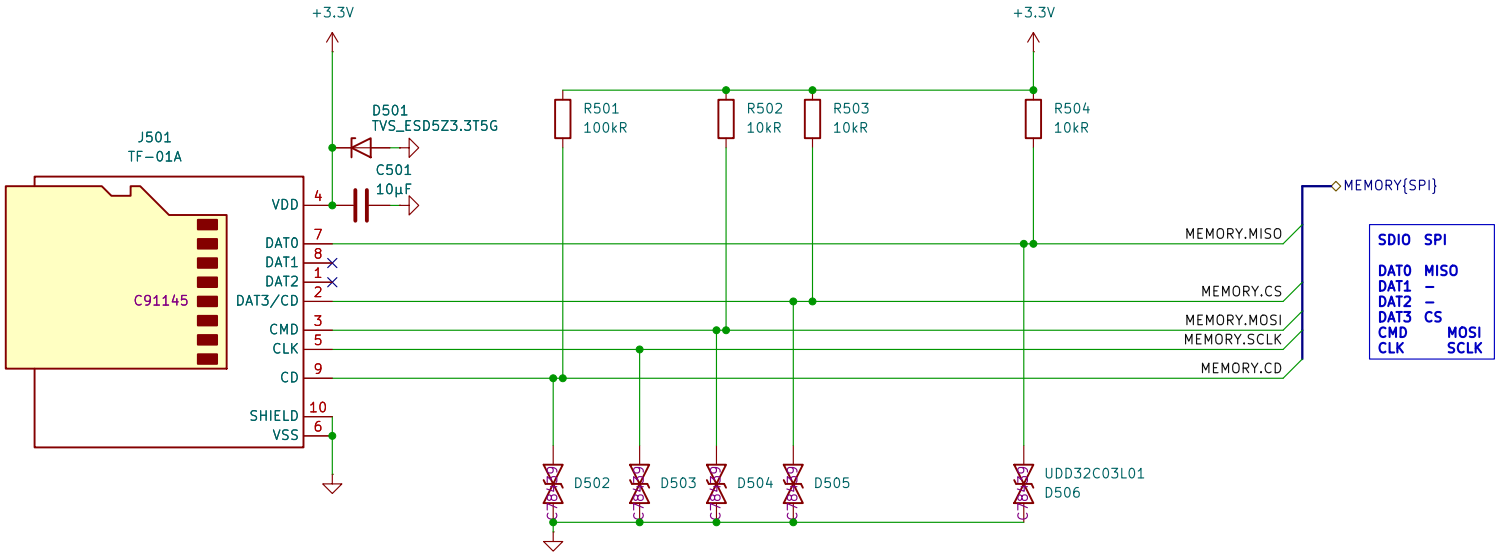
Title: BB_RPI_shield

Size: A4	Date: 2024-09-16	**Rev: v1**
KiCad E.D.A. 10.0.1		Id: 1/6


Quad-SPI FLASH NOR Memory
MCU
RP2040
U203
TAG-CONNECT
TC2050-IDC
TAG-CONNECT
J204
W25Q128JVS
U201
BNC
J202
74LVC1T45
U202
TVS_ESD5Z5.0T1G
D202
Trace impedance 50R 10%
matched length <120mm
CLK clearance 3x width
Debug the microcontroller using
the tag-connect:
TC2050-IDC + TC2050-ARM2010
with Raspberry Pi probe or J-link
module ID: | voltage:
RPI shield | 0V
screen | 0.4V
feeder | 0.8V
tunnel | 1.2V
lighting | 1.6V
nosepoke | 2V
RnD Unit
rnd@icm-institute.org
ICM – Paris Brain Institute
Sheet: /MCU/
File: mcu.kicad_sch
Title: BB_RPI_shield
Size: A4
Date: 2024-09-16
Rev: v1
KiCad E.D.A. 10.0.1
Id: 2/6


+12V
+5V
D301
TVS_ESD5Z5.0T1G
D302
TVS_ESD5Z5.0T1G
D303
TVS_ESD5Z12T1G
5 +5V
9 GND
4
8
3 GND
7 +5V
2 GND
6 +12V
1 +12V
J301
D09P33E4PA00LF
R303
60.4R
R304
60.4R
C301
4.7nF
+5V 1
GND 6
CANL 2
CANH 7
GND 3
+5V 8
GND 4
+12V 9
+12V 5
J302
D09S33E4PA00LF
TP303
TP304
The CAN controller + MCU parts alone consume up to 100mA
+3.3V
+5V
C303
100nF
C304
100nF
C305
100nF
C302
47pF
C306
4.7nF
C307
47pF
R305
60.4R
R310
60.4R
R307
10k
R308
10k
R309
10k
10k
R306
TP305
TP306
TP307
TP308
D304
D305
CAN{SPI1+GPIO}
CAN.STANDBY
CAN.SCK
CAN.MOSI
CAN.MISO
CAN.CS
CAN.RESET
CAN.TX0_RST
CAN.RX0_BF
CAN.INTERRUPT
STBY
SCK
SI
SO
CS
RESET
TX0RTS
TX1RTS
TX2RTS
RX0BF
RX1BF
INT
CLKOUT
VDD
VIO
VDDA
GND
VSS
CANH
CANL
RXD
TXD
TxCAN
RxCAN
OSC2
OSC1
Y301
U301
MCP25625-E/SS
RnD Unit
rnd@icm-institute.org
ICM – Paris Brain Institute
Sheet: /CAN bus/
File: CAN_bus.kicad_sch
Title: BB_RPI_shield
Size: A4
Date: 2024-09-16
Rev: v1
KiCad E.D.A. 10.0.1
Id: 3/6


USB-C
5V 4A buck converter
J403
PWR_FLAG
R409
5.1k
R410
5.1k
D407
TVS_ESD5Z5.0T1G
C411
2.2uF
trace impedance: 90Ohm
Same length +-10mm
R411
27R
R412
27R
D403
D404
TVS_ESD321DYAR
C408
47pF
C409
47pF
C412
100nF
U405
FT234XD
USB_detect
R413
470R
D402
LG401
515-1049F
UART_TX
UART_RX
UART_RTS
UART_CTS
J401
PWR_FLAG
R401
5.1k
R402
5.1k
D408
TVS_ESD5Z5.0T1G
C410
2.2uF
R404
27R
R405
27R
place close to mcu
USB_D-
USB_D+
D405
D406
TVS_ESD321DYAR
U401
LM66100DCK
C401
1uF
+3.3V
TP401
U402
MIC5504-3.3YM5
C402
1uF
C403
1uF
+12V
PWR_FLAG
TP403
J402
Barrel_Jack_Switch
C5441017
D401
TVS_ESD5Z12T1G
U404
ST1S41PHR
C2070789
L401
Ind_XAL5030-332MEC
C406
10uF 25V
C407
1uF 25V
R407
115k
R408
22k
+5V
TP402
PWR_FLAG
C405
22uF
U403
LM66100DCK
C404
1uF
TP404
GND
RnD Unit
rnd@icm-institute.org
ICM - Paris Brain Institute
Sheet: /Power/
File: power.kicad_sch
Title: BB_RPI_shield
Size: A4
Date: 2024-09-16
Rev: v1
KiCad E.D.A. 10.0.1
Id: 4/6



https://jlcpcb.com/partdetail/Korean_HropartsElec-TF01A/C91145

Micro SD card connector

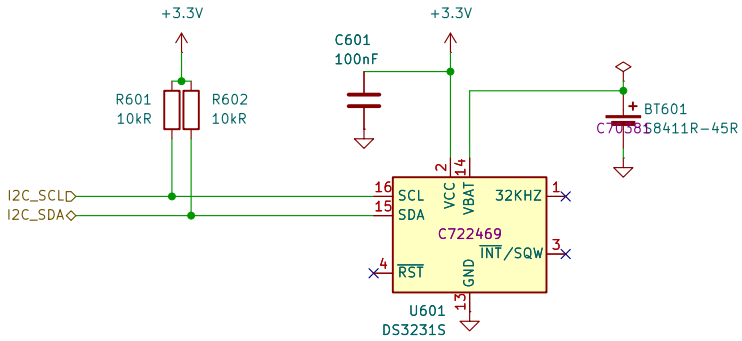
ESD & EMI filter

RnD Unit
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Sheet: /SD memory/
File: memory.kicad_sch

Title: BB_RPI_shield

Size: A4	Date: 2024-09-16	**Rev: v1**
KiCad E.D.A. 10.0.1		Id: 6/6



RTC

RnD Unit
rnd@icm-institute.org
ICM – Paris Brain Institute

Sheet: /RTC/
File: rtc.kicad_sch

Title: BB_RPI_shield

Size: A4	Date: 2024-09-16	**Rev: v1**
KiCad E.D.A. 10.0.1		Id: 7/6